Smart SDN – Experiment Report

BURST Run (run #13, auto mode)
Started 2026-11-05 06:24:50 · duration 200.8 s · status completed

Summary

Metric	Value
Automatic route changes	6
Average / peak slow-link throughput	2.67 / 5.00 Mbps
Sync RTT while on SLOW path	97.0 ms
Sync RTT while on FAST path (after reroute)	13.4 ms
RTT improvement after reroute	86.2 %
Synchronization packet loss (timeouts)	0.7 %

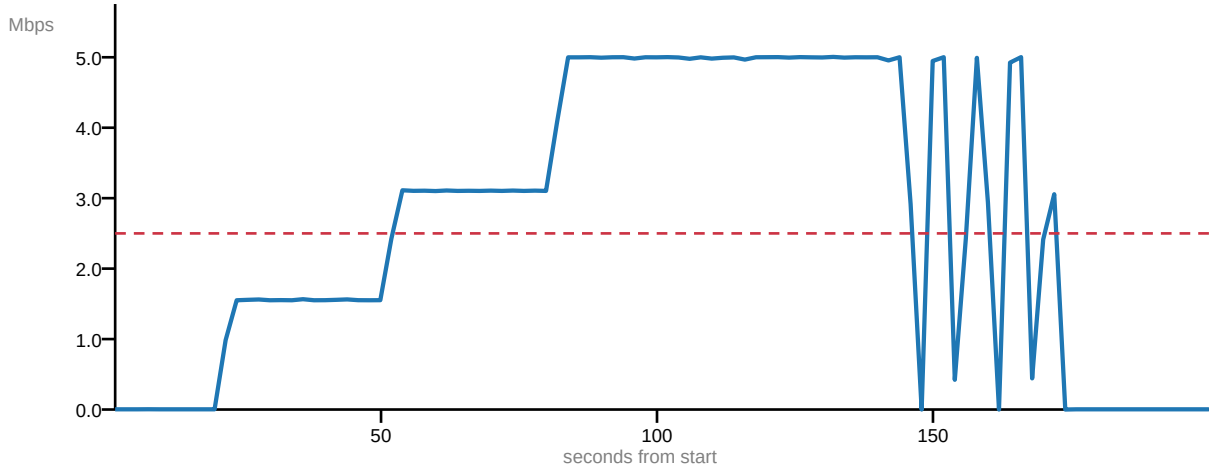
Per-phase results

Phase	Start (s)	Length (s)	Avg slow Mbps	Peak Mbps	Avg RTT ms	Loss %	SLOW/FAST samples
Baseline	1.8	20.0	0.00	0.00	101.3	0.0	10/0
Low	21.8	30.0	1.52	1.57	101.1	0.0	15/0
Moderate	51.9	30.0	3.06	3.11	20.0	0.0	1/14
High	81.9	30.0	4.93	5.00	10.6	0.0	0/15
Extreme	111.9	30.0	5.00	5.00	10.7	0.0	0/15
Burst	141.9	30.0	3.09	5.00	29.8	5.6	4/11
Recovery	172.0	30.0	0.21	3.06	88.2	0.0	14/1

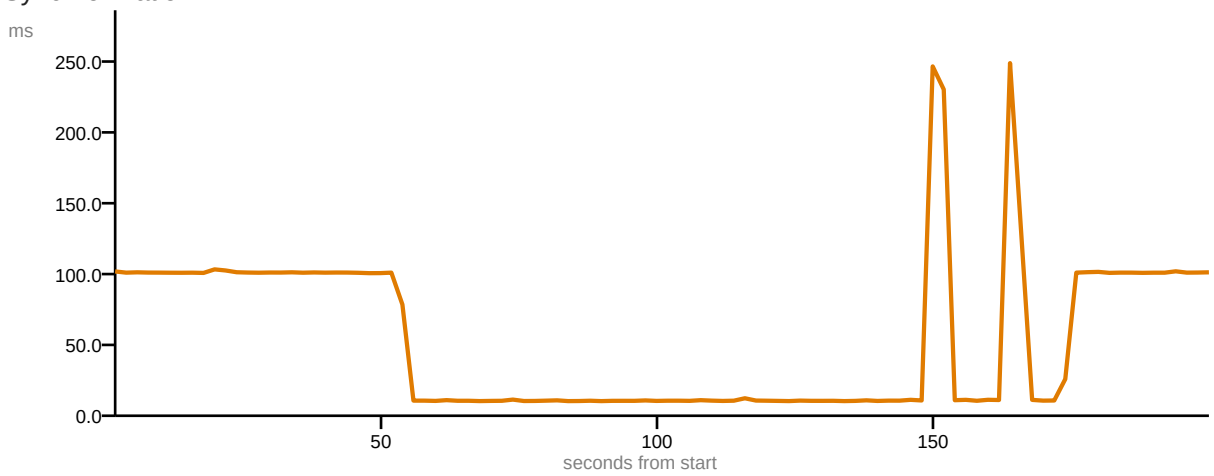
Controller routing events

Time	Phase	Event
2026-11-05 06:25:44	Moderate	SLOW -> FAST (slow link 3.11 Mbps)
2026-11-05 06:27:18	Burst	FAST -> SLOW (slow link 0.00 Mbps)
2026-11-05 06:27:22	Burst	SLOW -> FAST (slow link 5.00 Mbps)
2026-11-05 06:27:32	Burst	FAST -> SLOW (slow link 0.00 Mbps)
2026-11-05 06:27:36	Burst	SLOW -> FAST (slow link 5.00 Mbps)
2026-11-05 06:27:44	Recovery	FAST -> SLOW (slow link 0.00 Mbps)

Slow-link throughput (red dashed = reroute threshold)



Synchronization RTT



Generated 2026-11-05 06:28:16 by the Smart SDN dashboard. RTT values come from the UDP synchronization clients (h1–h3); throughput is measured by the Ryu controller on the slow link (s1 port 5).